ADS508 Data Science With Cloud Computing

Design Document

GlowSense Beauty Analytics

Team 1

Jee Hun Hwang | Lei Lin | Michelle Wang

ADS-508: Data Science With Cloud Computing

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GitHub: github.com/xuany823/ads508-FinalProject-Team1

Project Overview

Authors	Jee Hun Hwang, Lei Lin, Michelle Wang
Company Name	GlowSense Beauty Analytics
Company Industry	AI-Driven Retail Analytics for the Beauty Industry
Company Size	3
Course	ADS-508: Data Science With Cloud Computing
GitHub	https://github.com/xuany823/ads508-FinalProject-Team1

Abstract

GlowSense Beauty Analytics is a data science initiative that applied machine learning and natural language processing techniques to analyze product attributes and large-scale customer reviews from the Sephora online marketplace. The project examined factors influencing product success by modeling product ratings and analyzing customer sentiment. The resulting insights supported more informed decision-making in product assortment, pricing strategy, and marketing. By combining structured product data with unstructured review text, this study provided a comprehensive framework for understanding and predicting product performance in the beauty retail industry.

Problem Statement

The beauty industry introduces thousands of new products each year, yet many fail to achieve strong customer ratings or sustained market interest. Retailers operating on large online platforms such as Sephora must decide which products to promote, stock, and support with marketing resources despite limited early signals of success. This uncertainty creates risk, often resulting in inefficient inventory allocation and wasted marketing spend.

This project addressed that challenge by analyzing product attributes—including brand, pricing, and ingredient composition alongside early customer reviews. Predictive models were developed to estimate product success, defined by customer ratings and engagement metrics. These models enabled retailers to better identify high-potential products, reduce risk in product launches, and allocate resources more effectively.

Goals

- **Predictive Analytics:** Predict the likelihood that a new product on Sephora online store will achieve high customer ratings and strong popularity.
- **Prescriptive Analytics:** Identify the product attributes (e.g. price range, ingredients, brand factors, early customer review sentiment) that increase the chances of product success.
- **Informative:** Provide retailers with insights on product performance trends and customer preferences (based on love_counts or review volume) to support better product selection and marketing decisions.

Non-Goals

- **Product Development:** The project will not design or formulate new beauty products or recommend specific ingredient formulations.
- **Operational Decisions:** The system will not directly automate inventory purchasing, pricing, or marketing campaign execution for retailers.
- **Customer-Level Personalization:** The project will not build recommendation systems for individual shoppers or personalize product suggestions.

Data Sources

Source: https://www.kaggle.com/datasets/nadyinky/sephora-products-and-skincare-reviews

The analysis utilized a publicly available dataset from Kaggle titled Sephora Products and Skincare Reviews. The dataset included over 8,000 products and approximately one million customer reviews, containing structured attributes such as price, category, and ingredients, as well as unstructured review text.

Several risks were present when working with this dataset. Because the data was sourced from scraped online reviews, it contained missing values, inconsistent formatting, and duplicate records. In addition, the review text introduced a large volume of unstructured data (approximately one million reviews) which created computational challenges for natural language processing tasks. Another limitation was that product popularity metrics, such as ratings and review counts, may not fully represent true market success due to potential sampling bias among online reviewers.

The data was stored in an Amazon S3 data lake and processed using Amazon SageMaker. While the dataset was sufficiently large for machine learning applications, it presented several challenges, including missing values, inconsistent formatting, and potential bias in customer reviews. These limitations were addressed during data preparation to ensure robustness in modeling.

Data Exploration

1. Data Storage and Ingestion Strategy

The datasets were stored in an Amazon S3 bucket (ads508-team1-sephora) using a structured data lake architecture. Raw CSV files containing product and review data were uploaded to the raw layer to preserve the original data and maintain data lineage.

Data ingestion was implemented in Amazon SageMaker Studio, where notebooks read data directly from S3, performed initial cleaning (such as removing index columns), and consolidated multiple review files. The processed data was then stored in the curated layer in Parquet format to improve storage efficiency and query performance.

External tables were created in Amazon Athena to enable direct querying of curated datasets from S3. This architecture supported scalable and cost-effective data processing while enabling efficient exploratory analysis without repeatedly accessing raw data.

S3 Bucket Structure: ads508-team1-sephora

- raw/products/product_info.csv
- raw/reviews/reviews_0-250.csv, reviews_250-500.csv, reviews_500-750.csv, reviews_750-1250.csv, reviews_1250-end.csv
- curated/products/products.parquet
- curated/reviews/reviews.parquet
- analytics/query_results/
- athena/staging/

2. Exploration Focus Areas

During the exploration phase, the analysis focused on data quality, completeness, and consistency across both datasets. The reviews dataset contained over one million observations, while the products dataset included approximately 8,000 records. Several variables exhibited substantial missingness, including helpfulness, review_title, and demographic attributes such as skin_tone and eye_color. In the products dataset, variables such as value_price_usd and variation_desc also showed high rates of missing values.

The analysis then examined distributions of key numerical variables, including rating, helpfulness, total_feedback_count, loves_count, and price_usd. As seen in Figures 1 and 2, ratings were heavily skewed toward higher values (4–5 stars), while engagement metrics followed long-tailed distributions, indicating that a small number of products received disproportionately high engagement.

Figure 1

Feature Distribution of Products

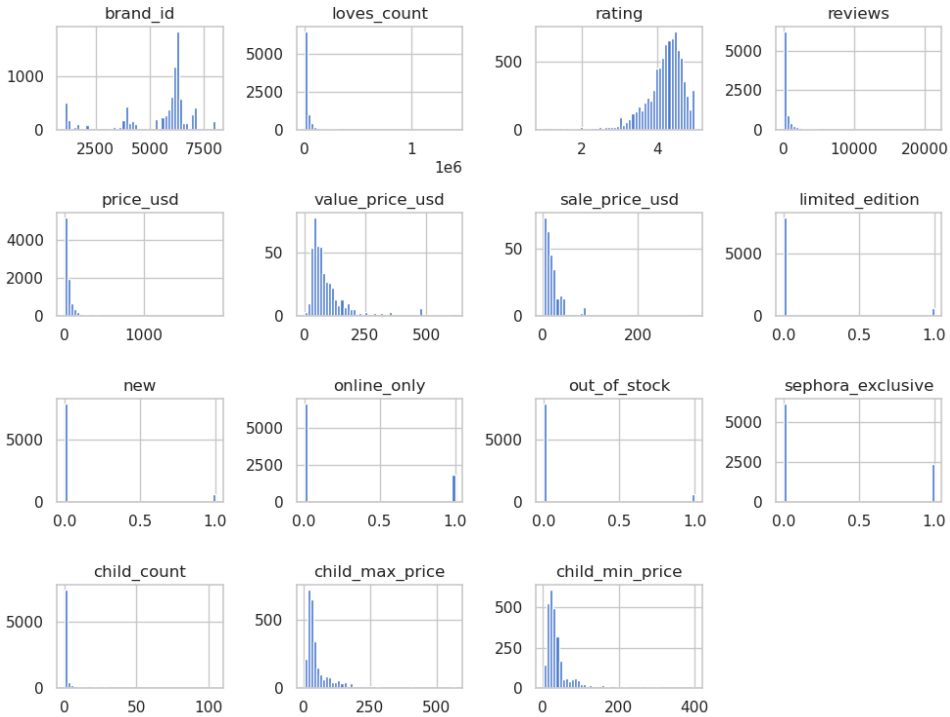
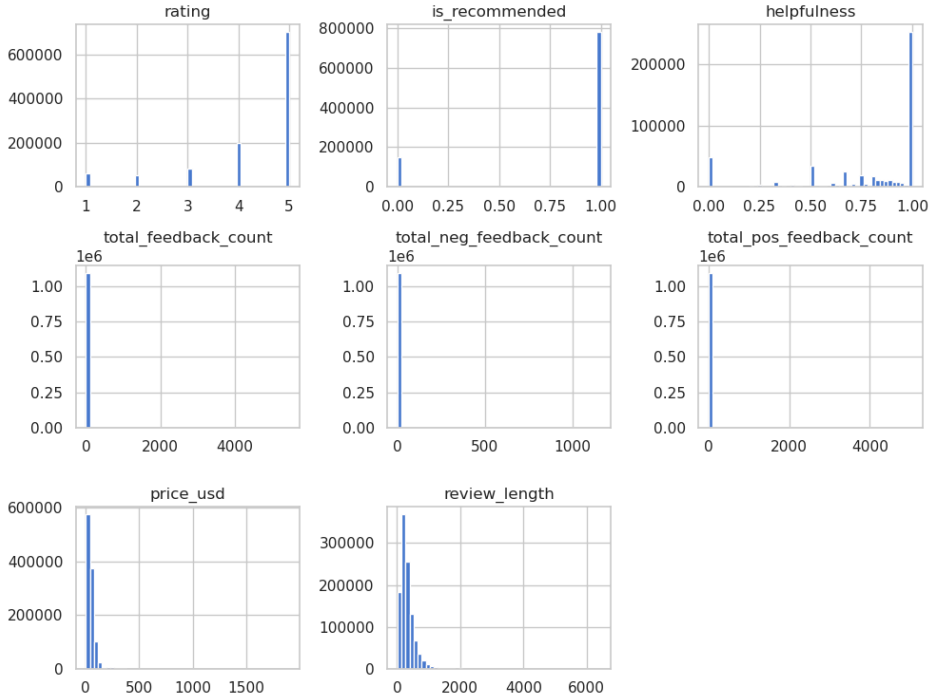


Figure 2

Feature Distribution of Reviews



Correlation analysis was conducted to identify relationships between variables and potential multicollinearity. Strong correlations were observed between rating and is_recommended, as well as between total_feedback_count and total_pos_feedback_count (Figure 3). In the products dataset, price-related variables such as price_usd, value_price_usd, and sale_price_usd were also highly correlated (Figure 4).

Figure 3

Correlation Heatmap of Reviews

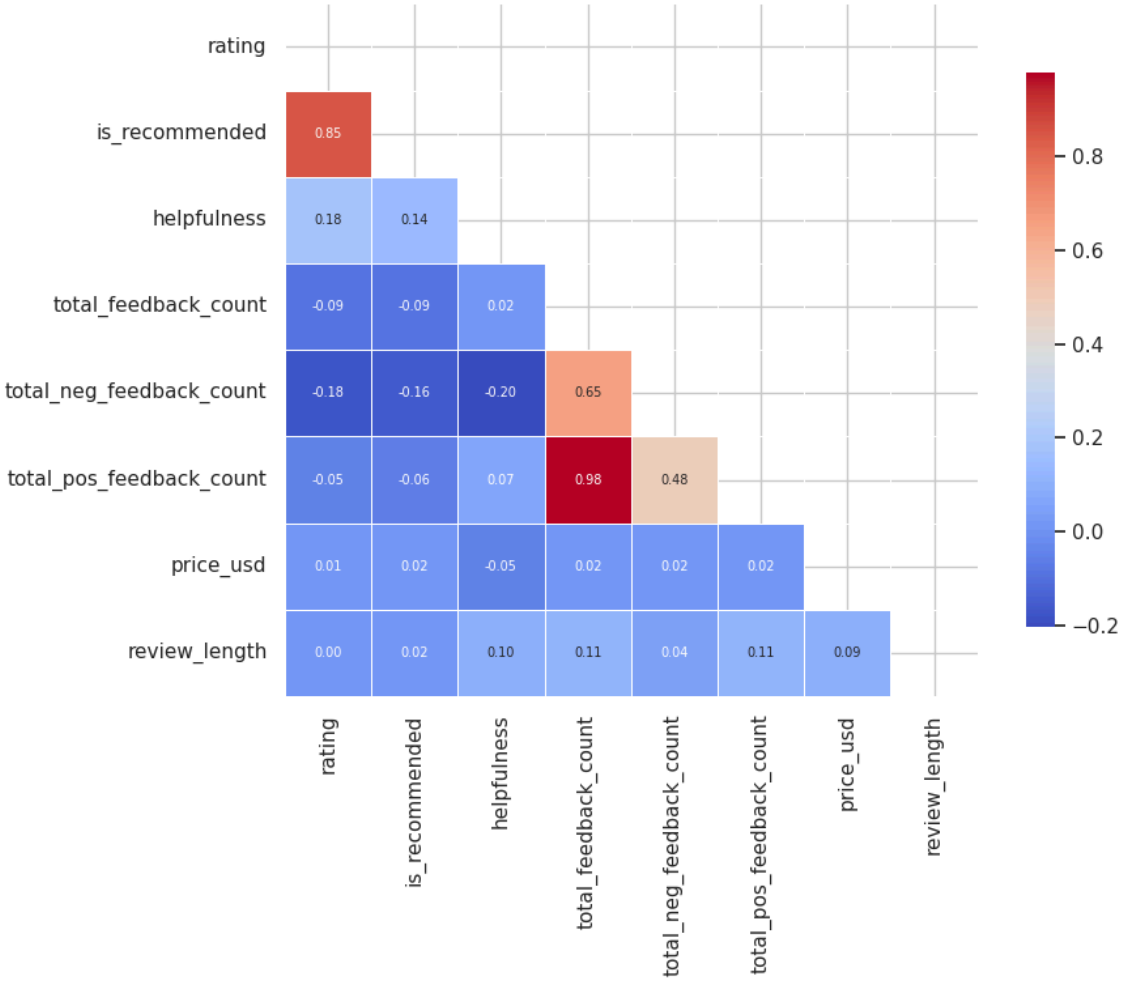
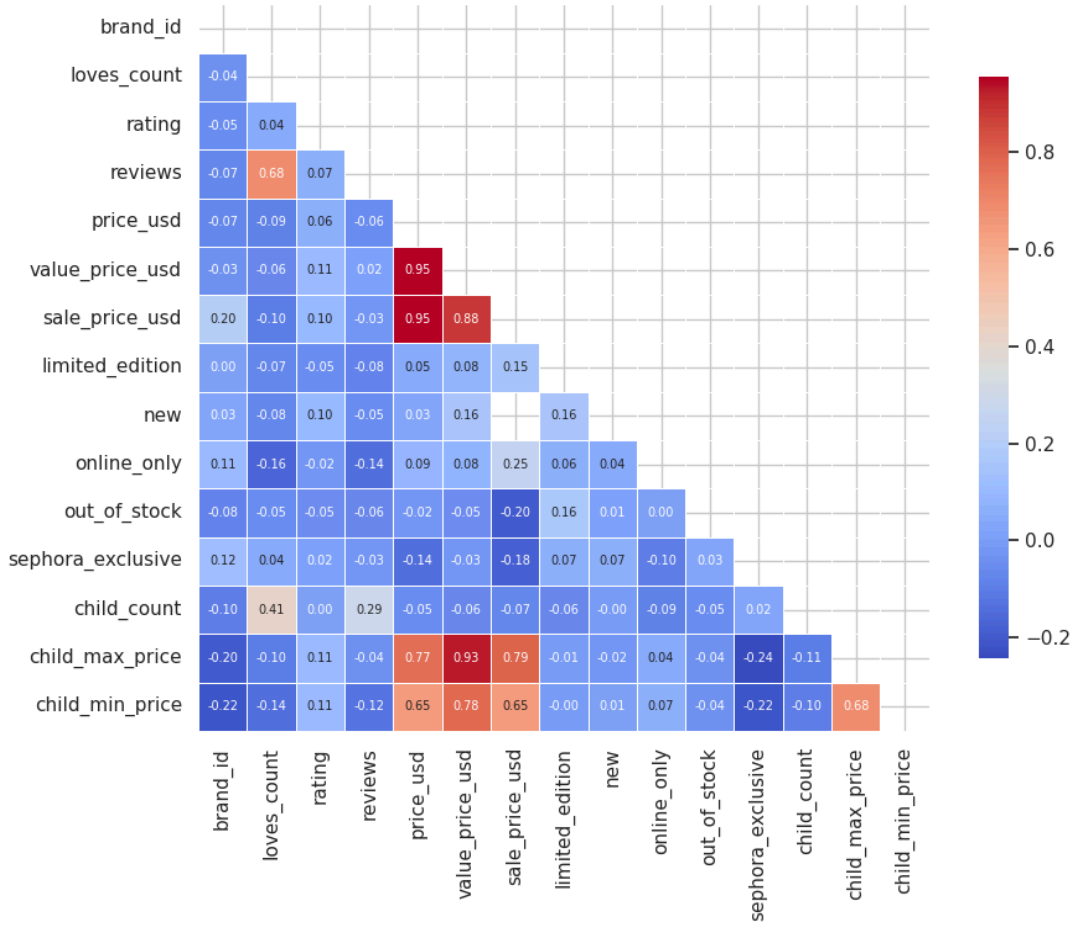


Figure 4

Correlation Heatmap of Products



Finally, the analysis considered potential sources of bias. Review data exhibited positivity bias, as customers were more likely to leave positive feedback. Additionally, popular products with high review counts or loves_count values tended to dominate the dataset, introducing popularity bias. Key predictive fields identified during exploration included product attributes (e.g., brand_name, price_usd, ingredients), engagement metrics (e.g., reviews, loves_count), and review_text for downstream NLP analysis.

3. Tools Used

- **Amazon S3:** data lake storage (raw and curated layers)
- **Amazon SageMaker Studio:** ingestion and EDA environment
- **Amazon Athena:** SQL-based querying of curated data
- **Python (pandas, pyarrow, seaborn, matplotlib):** data processing and visualization

4. GitHub Code Repository

Repository: https://github.com/xuany823/ads508-FinalProject-Team1/tree/main/Notebooks

Data Preparation

Data Scrubbing

The data cleaning process focused on handling missing values and refining the dataset to align with the project's business objective. The dataset was filtered to include only products within the Makeup category, reducing variability across fundamentally different product types and improving model consistency.

Missing values were handled using targeted imputation strategies. Numerical variables such as rating were imputed using the mean to preserve distributional characteristics. Count-based variables such as reviews were filled with zeros where missing values indicated absence. Categorical variables, including size, variation_type, and ingredients, were assigned a placeholder value ("NA") to retain observations while explicitly marking missing data.

For the reviews dataset, review_text and rating were retained as primary inputs. Because product identifiers were inconsistent across datasets, reviews were treated as an independent signal source rather than being directly joined with product data. Review text was transformed using TF-IDF with unigrams and bigrams, and VADER sentiment scores (positive, neutral, negative, compound) were added to capture overall tone.

Feature Selection

During feature selection, variables that did not contribute meaningful predictive value were removed. High-cardinality identifiers such as product_id, product_name, brand_id, and brand_name were excluded to prevent overfitting and noise. Raw text fields such as ingredients and highlights were also removed after transformation, as their information was already captured through engineered features.

Additionally, redundant variables were removed after transformation. For example, loves_count was replaced with a log-transformed version to reduce skewness. This process reduced dimensionality and ensured that the final feature set contained relevant, structured inputs for modeling.

Feature Creation

New features were engineered to capture key business signals related to product performance. A popularity index was created by combining rating and review count, identifying products that were both highly rated and widely reviewed.

Price was categorized into segments (budget, low_mid, mid, premium) to reflect consumer purchasing behavior. Ingredient-related features were derived by extracting the number of ingredients and applying TF-IDF to generate indicators for commonly occurring ingredients.

Feature Transformation

To prepare the dataset for modeling, several transformations were applied. Continuous variables such as price_usd and num_ingredients were normalized using standardization techniques to ensure consistent scale across features.

Categorical variables, including secondary_category and price_bucket, were converted into one-hot encoded representations. Log transformations were applied to skewed variables such as loves_count and price_usd to reduce the influence of extreme values and improve model stability.

Balancing the Dataset

To address class imbalance in product ratings, balancing techniques were applied during model training. Without balancing, models would disproportionately favor the most common rating categories.

Oversampling and class-weight adjustments were used to improve representation of underrepresented classes. These techniques were applied only to the training dataset, ensuring that validation and test sets remained representative of real-world distributions.

Dataset Split

The dataset was divided into training (90%), validation (5%), and test (5%) sets. The training set was used to fit the models, while the validation set supported hyperparameter tuning and model selection. The test set remained unseen during training and provided an unbiased evaluation of final model performance.

Model Training

SageMaker Approach

The project used a bring-your-own-script approach within Amazon SageMaker Studio. This approach allowed full control over preprocessing, feature engineering, and model configuration, which was necessary given the complexity of the engineered feature space.

Algorithms

Review-Based Model - SGDClassifier: An SGDClassifier was implemented due to its efficiency in handling high-dimensional, sparse TF-IDF features. This model scaled well to large datasets and performed effectively in classification tasks involving text data.

Product-Based Model -XGBoost Regressor: An XGBoost regressor was used. This model captured non-linear relationships between features such as price, ingredients, and product attributes, making it well-suited for predicting product performance.

Parameters

SGDClassifier: The following tuned parameters were used:

- loss = "hinge" (linear SVM objective for strong classification performance)
- alpha = 1e-4 (regularization strength to prevent overfitting)
- max_iter = 1000 and tol = 1e-3 (ensure convergence)
- class_weight = None (performed better than balanced in this case)

These parameters allow the model to efficiently learn from high-dimensional TF-IDF features while maintaining generalization.

XGBoost Regressor: The model was tuned using RandomizedSearchCV to maintain computational costs, resulting in:

- n_estimators = 250
- max_depth = 5
- learning_rate = 0.03
- subsample = 0.8
- colsample_bytree = 0.8
- reg_lambda = 1.5 (L2 regularization)
- reg_alpha = 0.05 (L1 regularization)

These parameters balanced bias and variance, controlled overfitting, and improved generalization. The relatively shallow depth and lower learning rate helped stabilize learning, while subsampling introduced randomness that improved robustness.

Instance Size

Model training was conducted on a single ml.m5.xlarge instance. This configuration was sufficient for the dataset size and computational requirements, as the models were CPU-based and did not require GPU acceleration. The selected instance balanced performance and cost efficiency.

Model Evaluation

Classification Metrics: Classification models were evaluated using accuracy, macro F1-score, and confusion matrices. Accuracy provided an overall measure of performance, while macro F1 ensured

balanced evaluation across all classes. As seen in Table 1, the SGDClassifier achieved an accuracy of 72.8% and a macro F1 score of 0.50, indicating strong performance on the classification task.

Table 1

Review-Based Model Results

Model	Accuracy	F1 (Macro)
SGDClassifier	0.728	0.503
Logistic Regression	0.714	0.477
Linear SVM	0.684	0.490

Regression Metrics: Regression models were evaluated using RMSE and R^2. The XGBoost model achieved the best performance, with an R^2 of approximately 0.37 and an RMSE of 0.47, as seen in Table 2. These results indicated that the model captured a meaningful portion of variation in product ratings while maintaining reasonable prediction error.

Table 2

Product-Based Model Results

Model	R^2	RMSE
XGBoost	0.368	0.470
Random Forest	0.322	0.487
Linear Regression	0.240	0.515

Feature Importance and Product-Level Insights

Feature importance analysis indicated that product success was influenced by a combination of pricing strategy, category positioning, ingredient composition, and customer engagement.

The interaction between price and review volume emerged as a key predictor, suggesting that products must balance affordability with popularity. Category features, particularly eye, cheek, and lip products, were consistently associated with higher ratings.

Ingredient-related features, including both individual ingredients and latent components derived from dimensionality reduction, contributed significantly to model performance. Additionally, factors such as online availability and product exclusivity influenced customer perception and ratings.

Review Text Insights and Customer Sentiment

Analysis of review text revealed clear patterns across rating categories. High ratings were associated with strong positive language, while mid-tier ratings often included mild criticism or price concerns. Low ratings were characterized by negative sentiment and references to dissatisfaction or adverse effects.

These findings indicated that customer ratings were closely tied to perceived effectiveness, value, and user experience. Negative reviews were frequently driven by unmet expectations or undesirable outcomes.

Measuring Impact

The success of the project was evaluated using both model performance metrics and business impact indicators.

1. **Prediction Accuracy for Product Success:** Measured using model evaluation metrics such as accuracy, precision, recall, and F1-score when predicting whether a product will achieve high customer ratings (e.g., rating ≥ 4.0) or strong popularity (e.g., high review count or loves_count). These metrics ensured that the predictive model reliably identified products with strong customer reception.
2. **Product Selection Decision Quality:** Evaluated how well the model identified key attributes associated with successful products. Feature importance analysis and model explainability techniques were used to identify influential variables such as price range, brand reputation, ingredient composition, and early review sentiment.
3. **Customer Sentiment Prediction Performance:** Monitored NLP-based prediction performance on review text. Metrics such as sentiment classification accuracy and average sentiment score distribution helped assess how effectively the system captures customer opinions about skincare products.

Actionable Next Steps

Building on these findings, several strategic actions can be taken to improve product performance and customer satisfaction. Pricing strategies should be carefully optimized to align perceived value with customer expectations, particularly given the strong influence of price-demand dynamics and the frequent mention of cost concerns in reviews. Product development efforts should prioritize high-performing categories such as eye, lip, and cheek products while incorporating ingredients associated with positive outcomes, such as hydrating and smoothing agents. At the same time, formulations should be refined to minimize the risk of irritation or adverse reactions, which are strongly associated with low ratings. Marketing strategies can be enhanced by incorporating language that resonates with customers, leveraging commonly used positive descriptors identified in high-rated reviews. Additionally, improving the integration between product attributes and customer feedback will enable more comprehensive insights across the product lifecycle, supporting both pre-launch predictions and post-launch evaluations. Finally, better customer targeting and clearer communication of product benefits, expected results, and suitability can help reduce mismatched expectations and improve overall satisfaction.

In summary, the regression-based product model and the classification-based review model provide complementary perspectives on product success. The product model enables forward-looking predictions based on structured attributes, while the review model captures real-world customer sentiment and experience. Together, these approaches offer a comprehensive framework for understanding and predicting product performance, supporting more effective decision-making in product development, pricing, and marketing.

Security Checklist, Privacy, and Other Risks

PHI Data

No. The dataset contains publicly available product information and customer review data related to skincare products. It does not contain any protected health information (PHI).

PII Data

No direct personally identifiable information (PII) is included. The dataset contains anonymized identifiers such as author_id, but these values do not allow identification of individual users.

User Behavior Tracking

Yes, but only indirectly through existing dataset fields. The dataset includes behavioral indicators such as ratings, recommendation flags, helpful votes, and review text, which represent customer engagement with products. However, this project does not collect new behavioral data, and it only analyzes historical public review data.

Credit Card Data

No. The dataset contains product and review information only and does not include payment or financial transaction data.

S3 Bucket Access

The application reads from and writes to a single Amazon S3 bucket: ads508-team1-sephora.

- **Read from:** s3://ads508-team1-sephora/raw/ (contains original product and review CSV files)
- **Write to:** s3://ads508-team1-sephora/curated/ (stores cleaned and processed datasets in Parquet format)
- **Optional output:** s3://ads508-team1-sephora/athena/staging/ (stores query results and temporary files)

Data Bias Considerations

- **Review Participation Bias:** Customers who leave reviews may represent individuals with very positive or very negative experiences, which may not represent the overall customer population.
- **Brand Popularity Bias:** Well-known brands may receive more reviews and higher engagement, which may skew popularity metrics such as loves_count or review volume.
- **Product Category Bias:** Certain product categories (e.g., skincare or fragrance) may have more reviews than others, leading to uneven representation across categories.
- **Demographic Bias in Reviews:** Some demographic attributes (e.g., skin tone, skin type) appear in the dataset but may be incomplete or inconsistently reported, which may affect downstream modeling if not handled carefully.

Ethical Concerns

Yes, several ethical considerations should be acknowledged.

First, the dataset contains user-generated content, including review text and demographic attributes such as skin tone or skin type. Care must be taken to ensure that models do not unintentionally reinforce demographic biases when analyzing product performance.

Second, predictive models could influence which products receive marketing support or shelf placement. If not carefully evaluated, the model may favor already popular brands or products, potentially harming smaller or newer brands.

Finally, transparency is important when using machine learning to inform business decisions. Retailers should treat the model's predictions as decision-support tools rather than automated decision-makers, ensuring that human judgment remains part of the final product selection process.

Future Enhancements

The following enhancements would improve the model pipeline with additional time and resources:

1. Advanced Feature Engineering with Deep Text & Ingredient Representations

One impactful improvement would be replacing TF-IDF + SVD with transformer-based embeddings (e.g., BERT) for ingredients and review text. While TF-IDF captures word frequency, it misses semantic meaning and contextual relationships between ingredients and review phrases. Using pretrained language models would generate richer embeddings that better capture nuances like ingredient interactions and sentiment, likely improving predictive performance given the nonlinear patterns observed in the XGBoost model.

Additionally, grouping ingredients into functional categories (e.g., emollients, preservatives, actives) would reduce dimensionality while improving interpretability. Combining domain knowledge with learned embeddings would strengthen signal quality and help the model generalize across similar products, leading to more actionable insights for product development.

2. Improved Data Integration Between Products and Reviews

A key limitation in the current pipeline is the inability to reliably join products and reviews due to inconsistent product_id values. With more time, implementing a matching system using product names, brands, and similarity scoring would enable better dataset integration. This would allow for unified feature sets combining structured product attributes with unstructured customer feedback like sentiment and key phrases.

This integration is impactful because it connects what a product is with how customers perceive it. For example, linking ingredients with review sentiment could reveal drivers of positive or negative experiences. From a business perspective, this would provide a more complete view of product success, improving decisions around product design, positioning, and marketing strategies.

3. Model Optimization, Ensembling, and Advanced Validation Strategies

Another major enhancement would be expanding and enriching the product-level dataset, which is currently limited in size and constrains the model's ability to learn meaningful patterns. With only around two thousand product records, the model struggles to capture complex relationships between features like price, ingredients, and category, leading to lower predictive power and higher variability.

By incorporating additional data, the model would gain stronger, more consistent signals. This expansion would significantly improve generalization and stability, allowing for more accurate predictions of product success. From a business perspective, a richer dataset would directly enhance decision-making by enabling more reliable insights into which product characteristics drive higher ratings, ultimately supporting better product development, pricing, and launch strategies in the Sephora marketplace.

End of Document

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